ALEXNET CODE

```
import tensorflow as tf
from tensorflow.keras.datasets import mnist
from tensorflow keras.models import Sequential
from tensorflow keras.layers import Conv2D, MaxPool2D,
    Flatten, Dense
from tensorflow keras.optimizers import Adam
from tensorflow.keras.utils import to_categorical, plot_model
(train_x, train_y), (test_x, test_y) = mnist.load_data()
print("The size of train_x is: {}".format(train_x.shape))
print("The size of train_y is: {}".format(train_y.shape))
print("The size of test_x is: {}".format(test_x.shape))
print("The size of test_y is: {}".format(test_y.shape))
train_x = train_x.reshape(train_x.shape[0], 28, 28, 1)
test_x = test_x.reshape(test_x.shape[0], 28, 28, 1)
train_x = train_x/255.0
test_x = test_x/255.0
train_y = to_categorical(train_y, num_classes=10)
test_y = to_categorical(test_y, num_classes=10)
model = Sequential()
model.add(Conv2D(96, (5,5), strides=4, padding='valid',
    input_shape=(28, 28, 1), activation='relu'))
model.add(MaxPool2D((2,2), strides=2))
model.add(Conv2D(256, (3,3), padding='same',
    activation='relu'))
model.add(MaxPool2D(2,2), strides=1))
model.add(Conv2D(384, (1,1), padding='same', activation='relu'
model.add(Conv2D(384, (1,1), padding='same', activation='re
model.add(Conv2D(256, (1,1), padding='same', activation='re
model.add(Flatten())
model.add(Dense(512, activation='relu'))
model.add(Dense(512, activation='relu'))
model.add(Dense(10, activation='softmax'))
model.summary()
model.compile(loss='categorical_crossentropy', optimizer=Adam
```

```
metrics= ['accuracy'])
test_loss_list =[ ]
test_accuracy_list =[ ]
for i in range (20):
    print ("\nIteration:", i+1)
        model.fit (train_x, train_y, batch_size = 128, epochs = 1
                verbose = 1  validation_data = (test_x, test_y))
    score = model.evaluate (test_x, test_y, verbose=0)
            print ("Test loss:", score[0])
            print ("Test accuracy:", score[1])
            test_loss_list.append (score[0])
            test_accuracy_list.append (score[1])
    print ("\nTest Loss for 20 iterations:", test_loss_list )
    print ("Test accuracy for 20 iterations:" test_accuracy_list)
```

Output:

```
The size of train_x is : (60000, 28, 28)
The size of train_y is : (60000,)
The size of test_x is : (10000, 28, 28)
The size of test_y is : (10000,)
Model "sequential"
Layer (type)                    output shape          Param#
conv2d (Conv2D)                 (none, 6, 6, 96)      2,496
max_pooling2d (MaxPool2D)       (none, 3, 3, 96)      0
Conv2d_1 (Conv2D)               (none, 3, 3, 256)     221,440
max_pooling2d_1 (MaxPool2D)     (none, 2, 2, 256)     0
Conv2d_2 (Conv2D)               (none, 2, 2, 384)     98,688
Conv2d_3 (Conv2D)               (none, 2, 2, 384)     147,840
Conv2d_4 (Conv2D)               (none, 2, 2, 256)     98,566
flatten (Flatten)               (none, 1024)          0
dense_1 (Dense)                 (none, 512)           262,656
dense (Dense)                   (none, 512)           524,800
dense_2 (Dense)                 (none, 10)            5,130
Total params : 1,361,610 (5.19MB)
```

Trainable params : 1,361,610 (5.19 MB)

non-Trainable Params : 0 (0.00B)

Iteration 1:

469/469

Test loss: 0.117774426937103 27

Test accuracy: 0.9621000289916992

Iteration 2:

469/469

Test loss: 0.09969805181026459

Test accuracy: 0.9696000218391418

Iteration 3:

469/469

Test loss: 0.8533988893032074

Test accuracy: 0.9732000231742859

Iteration 4:

469/469

Test loss: 0.07271480560302734

Test accuracy: 0.9787999987602234

Iteration 5:

469/469

Test loss: 0.6910333037376404

Test accuracy: 0.9779999852180481

Iteration 6:

469/469

Test loss: 0.13360665738582 61

Test accuracy: 0.9616006056266785

Iteration 7:

469/469

Test loss: 0.08006885647773743

Test accuracy: 0.9764999747276366

Iteration 8:

469/469

Test loss: 0.08264190703630447

Test accuracy: 0.9779999852180481

Iteration 9:
469/469
Test loss: 0.06705765753984451
Test accuracy: 0.9781000018119812

Iteration 10:
469/469
Test loss: 0.09154238551855087
Test accuracy: 0.9761999845504761